Q5 NO CALCULATOR ALLOWED Q5

Answer QUESTION 5 PARTS A and B on this page.

PART A

$\frac{dy}{dx} = (3-x)(y^2)$

$\int y^{-2}\,dy = \int (3-x)\,dx$

$-\frac{1}{3}y^{-3} = -1 + C$

$-\frac{1}{3y^3} = -1 + C$

$\frac{-1}{3(-1)^3} = -1 + C$

$\frac{-1}{-3} = -1 + C$

$\frac{1}{3} + 1 = C$

$C = \frac{4}{3}$

$-\frac{1}{3y^3} = -1 + \frac{4}{3}$

$\frac{1}{3y^3} = 1 - \frac{4}{3}$

$\frac{1}{y^3} = 3 - 4$

$\frac{1}{y^3} = -1$

$1 = -y^3$

$-1 = y^3 \quad y = -1$

$f''(1) = -1$

?

PART B

$\frac{f'(x)(x-c)^n}{n!}$

$f(1) = -1$

$f'(1) =$

$f''(1) = -1$

Use a pencil or a pen with black or dark blue ink. Do NOT write your name. Do NOT write outside the box.

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Q5525/12

Q5 ☒ NO CALCULATOR ALLOWED Q5

Answer QUESTION 5 PARTS C and D on this page.

PART C

$$\frac{f(z)(x-c)^{n+1}}{(n+1)!} \rightarrow \frac{(60)(0.1)^3}{3!} \leq 0.01$$

$$\frac{\cancel{60}(0.001)}{\cancel{6}}$$

$$10(0.001) \leq 0.01$$

$$0.01 \leq 0.01$$

0.1
0.1
01
000
0.01
0.1
001
0000
0.001

PART D $\Delta x = 0.2$ $\frac{dy}{dx} = (3-x)(y^2)(dx)$

$(1,-1)$ $(3-1)((-1)^2)(0.2) = (2)(0.2) = 0.4$

$(1.2,-0.6)$ $(3-1.2)((-0.6)^2)(0.2) = (1.8)(0.36)(0.2)$

$(1.4,-0.470)$

$f(1.4) \approx -0.470$

3
0.6
0.6
.36
000

1.8
0.8
.38
00

1.8
0.36
0.36
216
1080
.1296

0.6000
.1296
4704

Use a pencil or a pen with black or dark blue ink. Do NOT write your name. Do NOT write outside the box.